

Python Math & Random Modules Cheat Sheet

Math Module

Function	Description	Example
<code>math.sqrt(x)</code>	Square root	<code>math.sqrt(25)</code> -> 5.0
<code>math.pow(x,y)</code>	Power	<code>math.pow(2,3)</code> -> 8.0
<code>math.factorial(x)</code>	Factorial	120
<code>math.pi</code>	Pi constant	3.14159
<code>math.ceil(x)</code>	Round up	4
<code>math.floor(x)</code>	Round down	3
<code>math.fabs(x)</code>	Absolute	8.0
<code>math.trunc(x)</code>	Remove decimals	7
<code>math.gcd(a,b)</code>	GCD	10
<code>math.lcm(a,b)</code>	LCM	12
<code>math.sin(x)</code>	Sine	<code>math.sin(0)</code>
<code>math.cos(x)</code>	Cosine	<code>math.cos(0)</code>
<code>math.tan(x)</code>	Tangent	<code>math.tan(0)</code>
<code>math.log(x)</code>	Natural log	<code>math.log(10)</code>
<code>math.log10(x)</code>	Log base 10	2
<code>math.exp(x)</code>	e^x	<code>math.exp(2)</code>
<code>math.degrees(x)</code>	Radians->Degrees	180
<code>math.radians(x)</code>	Degrees->Radians	3.14

Random Module

Function	Description	Example
<code>random.randint(a,b)</code>	Random integer	1..10
<code>random.randrange()</code>	Range	0,2,...18
<code>random.random()</code>	Float 0-1	0.42
<code>random.uniform(a,b)</code>	Random float	1-5
<code>random.choice(seq)</code>	Random item	<code>choice(colors)</code>
<code>random.choices(seq,k)</code>	Multiple items	k=3
<code>random.sample(seq,k)</code>	Unique items	<code>sample(list,2)</code>
<code>random.shuffle(seq)</code>	Shuffle list	<code>shuffle(list)</code>
<code>random.seed(x)</code>	Seed	<code>seed(5)</code>